

## ORIGINAL ARTICLE



# The protection afforded by Berberine against chemotherapy-mediated nephropathy in rats involves regulation of the antioxidant axis

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## Abstract

Doxorubicin (DOX) treatment in cancer patients leads to nephrotoxicity. The nephroprotective effect of Berberine (BBR), a herbal ingredient, is well documented as antioxidant and activation of the Nrf2 signalling. This study aimed to investigate if Nrf2 is a major protective mechanism of BBR in DOX animal models. Rats were divided as ( $n = 6$  each): Control, BBR (100 mg/kg, orally), DOX (15 mg/kg, orally), BBR + DOX, and BBR + DOX + brusatol (0.2 mg/kg, i.p./twice per week) (an Nrf2 inhibitor). DOX was given as a single dose (day 10), whereas BBR was administered for 3 weeks on a daily basis. BBR reduced tubular degeneration and improved renal markers in DOX-treated rats. It also reduced renal nuclear levels of NF- $\kappa$ B p65, total reactive oxygen species (ROS), lipid peroxides, interleukin-6 (IL-6), and tumour necrosis factor- $\alpha$  (TNF- $\alpha$ ), as well as mRNA levels of Bax and cleaved caspase-3. However, BBR stimulated glutathione (GSH) and superoxide dismutase (SOD) levels, the transcription of Bcl2, and the mRNA, total cytoplasmic, and nuclear levels of Nrf2 with no effect on the cytoplasmic Keap1 levels. All these effects disappeared by brusatol. In conclusion, BBR prevents DOX-induced renal damage by activating Nrf2.

## KEYWORDS

Berberine, doxorubicin, kidney, Nrf2, oxidative stress

## 1 | INTRODUCTION

The kidneys are among the most vital organs in our body that play a significant role in infiltrating blood, electrolyte balance, waste removal, controlling blood pressure, activating vitamin D, and red blood cell production.<sup>1</sup> Therefore, maintaining our kidney health remains a priority in all conditions. Chronic kidney disorders (CKDs) are diseases that are characterized by prolonged loss of renal function for more than 3 months and are leading

causes of end-stage renal disease (ESRD), increased mortality, and burden on the health system.<sup>2</sup> Currently, the global prevalence of CKDs is about 8% to 16%.<sup>2</sup> Clinical renal manifestation among patients with CKDs includes structural abnormalities, reduced glomerular filtration rate (eGFR), albuminuria, and microalbuminuria.<sup>1,3</sup> Although the aetiology of CKDs remains idiopathic mainly, the common risk factors for the development of CKDs include genetics (e.g., sickle cell trait), smoking, high blood pressure, cardiovascular disorders

(CVDs), diabetes mellitus (DM), and chemotherapy, environmental toxins, pesticides, and chemotherapeutic agents.<sup>4,5</sup>

Doxorubicin is a major drug for the treatment of solid tumours.<sup>6</sup> During the last few years, physicians have reduced the application of this drug due to its associated systemic toxicities, such as renal and hepatic toxicities.<sup>7</sup> Indeed, DOX-mediated nephrotoxicity and CKD are well established in cancer patients and animal models.<sup>8–10</sup> Currently, it is well established that DOX is able to induce tissue inflammation, apoptosis, and fibrosis through its ability to generate large quantities of reactive oxygen species (ROS), which entails its mechanism of action.<sup>11–13</sup> The mechanism by which DOX generates ROS and the adverse effect of these ROS within the cells are described in excellent studies.<sup>14–18</sup>

Several studies have also shown an essential role of the transcription factor, the nuclear factor erythroid 2-related factors 2 (Nrf2) in mediating DOX-associated cardiac, renal, and hepatic damage.<sup>19–21</sup> This was also identified as a leading mechanism for the induction of DOX-induced cellular oxidative stress in these tissues.<sup>19–21</sup> Indeed, the major function of Nrf2 in most cells is to stimulate the synthesis of cellular enzymatic and nonenzymatic antioxidant systems such as glutathione (GSH) and phase II antioxidant enzymes.<sup>22–24</sup> Also, Nrf2 is an effective anti-inflammatory and antiapoptotic factor that could upregulate Bcl2 and suppress the nuclear factor kappa beta (NF- $\kappa$ B).<sup>24–26</sup> In addition, Nrf2 exerted other functions, including the regulation of metabolism, unfolded protein response (UPR), mitochondria function, and autophagy.<sup>24</sup> Levels of Nrf2 are significantly depressed in the kidneys of DOX-treated animals and were associated with renal oxidative stress inflammation and apoptosis.<sup>27–29</sup> On the contrary, stimulating Nrf2 prevented renal damage by reversing all these damaging pathways.<sup>27–29</sup> Therefore, it seems reasonable that any drug stimulating Nrf2 will be an effective strategy to alleviate DOX-induced renal damage.

Berberine (BBR) is one of the most active alkaloids isolated from *Berberis vulgaris* and *Coptidis rhizome*.<sup>30,31</sup> Currently, multiple health benefits are described for BBR, including hypolipidaemic, hypoglycaemic, antidiabetic, and anticancer potentials.<sup>31,32</sup> Besides, the dietary intake of BBR protected against cardiac, renal, neural, and hepatic damage in several models due to its well-characterized pharmacological antioxidant and anti-inflammatory properties.<sup>33–36</sup> Having the kidney in particular, the nephroprotective effects of BBR was also shown in vitro and in vivo in numerous renal injury animal models.<sup>31,35,37–41</sup> Interestingly, the hepatic and neural protective effect of BBR against hyperglycaemia and diabetes mellitus (DM)-mediated renal injury, as well

as in other models such as fed high-fat diet (HFD) and carbon tetrachloride (CCl<sub>4</sub>) models.<sup>32–34,39,41,42</sup> In the same line, BBR attenuated hepatic steatosis and damage in rats by activating Nrf2. A similar mechanism of protection was shown in the neural cells exposed to oxidants.

However, new investigations have also confirmed the protective effects of BBR against cisplatin and DOX-induced renal damage.<sup>43,44</sup> However, the mechanism of the reno-protection of BBR in DOX-treated animals was not investigated enough and was limited to increasing antioxidant levels and suppressing lipid peroxidation.<sup>44</sup> Therefore, given its important stimulatory role on renal and systemic Nrf2 expression in the other above-mentioned animal models, in this study, we have sought to examine if BBR could prevent DOX-mediated renal injury by stimulating Nrf2/antioxidant axis.

## 2 | MATERIALS AND METHODS

### 2.1 | Animals

Adult male rats (Wistar) were used in this study (10 weeks old/ $210 \pm 18$  g). Rats were supplied by and maintained at the Experimental Animal Care Center, King Saud University (KSU) in Riyadh, Saudi Arabia. Standard fixed living conditions, including ( $23 \pm 1^\circ\text{C}$ ; dark/light cycle) were considered all through the adaption (1 week) and experimental periods. All protocols that were followed in this experiment were approved by the institutional review board (IRB) at Princess Nourah University, Riyadh, KSA (IRB Number 20-0391). Also, this study was conducted in accordance with the Basic & Clinical Pharmacology & Toxicology policy for experimental and clinical studies.<sup>45</sup>

### 2.2 | Drugs and preparation

Carboxymethylcellulose (CMC) (as universal solvent) (Cat. # C5013) and BBR (Cat. # 60010), DOX hydrochloride (# D1515), and brusatol (C26H32O11; Cat # SML1868), a drug that promotes Nrf2 ubiquitination and degradation (an Nrf2 inhibitor) were provided by Sigma Aldrich, MO, USA. BBR and brusatol were always dissolved in 5% CMC to the desired concentration used in the experimental procedure. The 5% CMC was used as the vehicle.

### 2.3 | Experimental design

The classifications of the groups of this study were as followed ( $n = 8/\text{group}$ ): (1) **Control rats**: received the

vehicle, 5% CMC for 3 weeks, daily; (2) **BBR-treated rats**: orally administered **BBR (200 mg/kg)** for 3 weeks, daily; (3) **DOX-model rats**: orally administered 5% CMC for 3 weeks, daily, and i.p. treated with DOX (20 mg/kg, a single dose) only on day 10. (4) **BBR + DOX-treated rats**: orally administered BBR (200 mg/kg) for 3 weeks, daily, but i.p. treated with DOX (20 mg/kg) only on day 10; and (5) **BBR + DOX + brusatol-treated rats**: orally administered BBR (200 mg/kg) for 3 weeks, daily, but i.p. co-treated with DOX (20 mg/kg; a single dose only on day 10) and brusatol (2 mg/kg; every 3 days). Body weights were reported weekly, whereas food intake was recorded daily.

## 2.4 | Selection of drug doses

The selected doses of DOX and BBR, as well as the design of the experiment were selected in accordance with the study of Fouad et al.,<sup>44</sup> which reported an antioxidant protection of BBR, at this dose, in DOX rat model. The in vivo administration of brusatol at this dose to block hepatic Nrf2 has been reported by Shatoor et al.<sup>46</sup>

## 2.5 | Urine collection and analysis

The collection of urine from all groups was conducted using metabolic cages over 24 h. Urine samples were measured using a graduated cylinder and then centrifuged (12 000×g/10 min) and stored at −20°C. The concentrations of albumin, urea, and creatinine (Cr) were measured by kits (Ab108789 Abcam, MA, USA), (Cat # 80340, Crystal Chem, Zaandam, Netherlands), and (# Ab56340, Abcam, MA), respectively. All analyses were performed for  $n = 8$  rats/groups and as instructed by each kit using the Spectra-Max Plus Microplate Reader (S/N: 02927 SoftMax Pro 5; Molecular Devices, CA, USA).

## 2.6 | Blood analysis and biochemical analysis

Blood samples were collected from all rats of all groups and centrifuged (1500×g/10 min) to collect serum. All samples were stored at −20°C and used later to measure levels of Cr and albumin using the same kits and plate reader used for their analysis in the urine and for  $n = 8$  rats/group (see above). The calculation of urine flow rate and Cr clearance (CrCL) and excretion (CrE) was adopted as described by Bazzano et al.<sup>47</sup>

## 2.7 | Tissue collection, homogenization, and measurements

Anaesthesia was conducted using a ketamine/Xylazine hydrochloride mixture, and euthanasia was introduced by neck dislocation. The kidneys were removed surgically and washed with ice-cold phosphate-buffered saline (PBS) (pH = 7.4). The kidney parts were either fixed in 10% buffered formalin (for histology) or stored at −80°C (for the biochemical analysis). Kidney homogenization was prepared from the frozen kidney sections by sonication in PBS and centrifugation (12 000×g) to collect supernatants. These supernatants were stored at −80°C and used later to measure levels of ROS (#ab238535; Abcam, Cambridge, UK), malondialdehyde (lipid peroxides/MDA) (Cat. # 10009055, Cayman, MI, USA), superoxide dismutase (SOD) (Cat. # MBS036924, MyBioSource, CA, USA), total glutathione (GSH) (Cat. # orb782371, Biorbit, MO, USA), tumour necrosis factor- $\alpha$  (TNF- $\alpha$ ) (Cat. # MBS2507393, MyBioSource, CA, USA), interleukin-6 (Cat. # R6000B, R&D System, MN, USA), and Nrf2 (# MBS752046, MyBioSource, CA, USA). Also, the preparation of the nuclear proteins from the frozen kidneys was performed by a commercial kit (# 78833, ThermoFisher Scientific, MA, USA) and used to measure the levels of NF- $\kappa$ B p5 (# MBS752046 and # MBS2505513, MyBioSource, CA, USA). Measurements were conducted in duplicate for  $n = 8$  sample/groups and as instructed by each kit, and the absorbance of the plates was measured and analysed using the above-mentioned Spectra-Max Plus Microplate Reader.

## 2.8 | Real-time PCR (qPCR)

Based on previously published reports, primer sequences for all selected targets were designed and provided from Qiagen (Germany).<sup>19,48,49</sup> All primers used for this part are shown in Table 1. The total RNA isolation from all frozen renal tissue was conducted by a special isolation kit (# 74004, Qiagen, Germany). The cDNA synthesis kit (# K1621, ThermoFisher Scientific) was used to synthesize the template cDNA for all isolated RNA. The amplification of all samples followed the available master mix kit (Ssofast Evergreen Supermix, # 172-5200, BioRad, USA) using the CFX 348 qPCR machine. This kit contains all required ingredients and a full description of all amplification conditions needed to amplify the template cDNA. In accordance, the following ingredients were included in the qPCR reaction (20  $\mu$ l): 10  $\mu$ l of the master mix reagent, 0.2  $\mu$ l of each primer (final concentration 500 nM), 2  $\mu$ l of the template (final concentration 50 ng), and 7.2  $\mu$ l nuclease-free water. Amplification

**TABLE 1** Primers used in the real-time polymerase reaction

| Gene      | Primers (5' → 3')                                             | GenBank accession # | Product length |
|-----------|---------------------------------------------------------------|---------------------|----------------|
| Nrf2      | F: -AAAATCATTAACCTCCCTGTTGAT<br>R: '-CGGCGACTTTATTTCTTACCTCTC | NM_031789           | 118            |
| Bcl2      | F: GTGGATGACTGAGTACCTGAAC<br>R: GAGACAGCCAGGAGAAATCAA         | U34964.1            | 125            |
| Bax       | F: F: GATGGCCTCCTTTCCTACTTC<br>R: CTTCTTCCAGATGGTGAGTGAG      | NM_017059           | 96             |
| Caspase-3 | F: -AATTCAAGGGACGGGTCATG<br>R: GCTTGTGCGCGTACAGTTTC           | U49930              | 67             |
| B-actin   | F: ACAGGATGCAGAAGGAGATTAC<br>R: ACAGTGAGGCCAGGATAGA           | NM_031144           | 117            |

steps were heating (1 cycle/98°C/30 s), denaturation (40 cycles/98°C/5 s), annealing (40 cycles/60°C/5 s), and melting (1 cycle/95°C/5 s/step). Normalization was conducted on the machine-associated software against the reference gene and using the  $\Delta\Delta^{\text{CT}}$  method. qPCR was conducted for  $n = 8$  samples/group.

## 2.9 | Immunoblotting

Proteins were extracted from frozen tissue using radio-immunoprecipitation (RIPA) buffer, and their levels were measured using the ThermoFisher Scientific protein assay kit (# 23225, USA). For each sample, proteins were prepared in the loading buffer (1 µg/µl) and then boiled for 5 min. All samples were then loaded (20 µg/well) on the gel and separated at 100 mV for 2 h using SDS-PAGE. Nitrocellulose membranes were used to wet transfer of the proteins at a voltage of 80 mV/2 h. Membranes were then blocked, washed, and incubated with the monoclonal antibodies against Nrf2 (# sc-365 949, 1:500, 60 kDa, Santa Cruz Biotechnology, USA), lamin A (# sc-518 013, 1: 1000, 69 kDa, Santa Cruz Biotechnology, USA), Keap1 (# 8047, 1:1000, 64 kDa, Cell Signalling Technology, USA), and  $\beta$ -actin (# 3700, 1:1000, 45 kDa, Cell Signalling Technology, USA). Each membrane was then washed with the TBST buffer (3×) and incubated with the 2nd antibody that corresponds to the selected primary for 2 h with shaking. In addition, washing between steps (three times/wash) and the first and second antibodies dilution was done using the 1× Tris-buffered Saline, 0.1% Tween-20 (TBST buffer). The band developments on each membrane were achieved by 5 min incubation with the enhanced chemiluminescence (# sc-2048, Santa Cruz Biotechnology, USA) and capturing the image using the C-Di Git blot scanner. Western blot was conducted in triplicate for each group of rats.

## 2.10 | Histopathological evaluation

All tissue were deparaffinized using xylene and alcohol (descending levels of 100%, 90%, and 70%, respectively). After being embedded in wax, all tissues were cut using a rotatory microtome at a thickness of 3–5 µm and then routinely stained with Harris haematoxylin (H) and Eosin (E). Photographing and capturing the images were done in a blind manner at 200×.

## 2.11 | Statistical analysis

All results were inserted and analysed using version 8 of the GraphPad Prism analysis package (Australia). Normality and analysis were conducted using the Kolmogorov–Smirnov and the two-way ANOVA tests, respectively. Tukey's test was used as post hoc ( $p < 0.05$ ). The expression of the data in the results was presented as means  $\pm$  standard deviation (SD).

## 3 | RESULTS

### 3.1 | Changes in body weights and markers of kidney function

Daily administration of BBR to control animals did not significantly change their kidney weights and food intake nor any of the measured kidney-related markers in the serum and urine (Table 2). Final body weights and average food intake were significantly decreased in DOX-treated rats as compared to the control (Table 2). Besides, DOX-treated rats showed higher Cr and urea serum levels with a parallel reduction in serum albumin levels. They also reduced daily urine production, urinary flow rate, and urinary Cr levels with a concomitant reduction

**TABLE 2** BBR ameliorates the loss in body weights and attenuates kidney function in DOX-treated rats

| Parameter                    | Control      | BBR          | DOX                         | BBR + DOX                    | BBR + DOX + brusatol           |
|------------------------------|--------------|--------------|-----------------------------|------------------------------|--------------------------------|
| Final body weights (g)       | 310.1 ± 12.3 | 307.3 ± 11.8 | 263 ± 10.6 <sup>a,b</sup>   | 299 ± 13.1 <sup>c</sup>      | 315 ± 16.1 <sup>a,b,c,d</sup>  |
| Weekly food intake (g/group) | 1593 ± 124.1 | 1702 ± 110.2 | 1023 ± 82.4 <sup>a,b</sup>  | 1573 ± 111.5 <sup>c</sup>    | 1489 ± 92.7 <sup>a,b,c,d</sup> |
| <b>Serum</b>                 |              |              |                             |                              |                                |
| Albumin (g/dl)               | 5.4 ± 1.1    | 5.7 ± 1.2    | 2.67 ± 0.7 <sup>a,b</sup>   | 4.91 ± 1.0 <sup>b,c</sup>    | 2.49 ± 0.6 <sup>a,b,d</sup>    |
| Urea (mg/dl)                 | 6.1 ± 1.34   | 6.4 ± 1.19   | 26.5 ± 5.3 <sup>a,b</sup>   | 9.1 ± 1.4 <sup>a,b,c</sup>   | 25.3 ± 6.1 <sup>a,b,d</sup>    |
| Creatinine (mg/dl)           | 0.75 ± 0.2   | 0.71 ± 0.1   | 2.87 ± 0.6 <sup>a,b</sup>   | 1.05 ± 0.3 <sup>a,b,c</sup>  | 2.69 ± 0.6 <sup>a,b,d</sup>    |
| <b>Urine</b>                 |              |              |                             |                              |                                |
| Volume (ml/24 h)             | 14.5 ± 2.7   | 13.8 ± 3.1   | 6.7 ± 1.6 <sup>a,b</sup>    | 12.1 ± 2.8 <sup>c</sup>      | 6.1 ± 1.7 <sup>a,b,d</sup>     |
| Urine flow (μl/min)          | 10.4 ± 1.7   | 9.5 ± 1.5    | 4.7 ± 1.1 <sup>a,b</sup>    | 8.4 ± 1.8 <sup>c</sup>       | 4.2 ± 1.0 <sup>a,b,d</sup>     |
| Albumin (μg/dl)              | 20.3 ± 3.5   | 18.9 ± 4.1   | 65.4 ± 5.6 <sup>a,b</sup>   | 27.5 ± 5.1 <sup>a,b,c</sup>  | 69.4 ± 7.4 <sup>a,b,d</sup>    |
| Creatinine (mg/dl)           | 68.4 ± 7.5   | 72.3 ± 6.2   | 24.3 ± 2.9 <sup>a,b</sup>   | 59.8 ± 5.2 <sup>a,b,c</sup>  | 27.1 ± 3.9 <sup>a,b,c</sup>    |
| CrCl (ml/min)                | 0.92 ± 0.12  | 0.97 ± 0.16  | 0.04 ± 0.003 <sup>a,b</sup> | 0.48 ± 0.08 <sup>a,b,c</sup> | 0.04 ± 0.003 <sup>a,b,d</sup>  |

Note: Data are given as mean ± SD (*n* = 6 animals/group).

<sup>a</sup>Versus control rats.

<sup>b</sup>Versus BBR-treated rats.

<sup>c</sup>Versus DOX-treated rats.

<sup>d</sup>Versus BBR + DOX-treated rats.

in urinary Cr clearance (CrCl) (Table 2). In addition, DOX-treated rats showed higher urinary albumin levels (Table 2). All these alterations in DOX-treated rats were significantly reversed in BBR + DOX-treated rats (Table 2). On the other hand, while there were no significant variations in the rats' final body weight and weekly food intake, serum levels of albumin and urea, urine volume, urine flow, and urinary Cr and CrCl were significantly decreased. Still, serum urea and Cr and urinary albumin levels were significantly increased in BBR + DOX + brusatol-treated rats as compared to BBR + DOX-treated animals. These levels were all not significantly varied as compared to DOX-treated rats (Table 2).

### 3.2 | Histological findings

Normal renal morphology with intact glomerulus, glomerular mass, and membranes, and proximal convoluted tubules (PCT) and distal convoluted tubules (DCT) were observed in the control and BBR-treated rats (Figure 1A, B). Kidneys from DOX-treated rats showed severe degeneration in the glomerular membranes and both the PCT and DCT and increased Bowman's capsule space. Many nuclei forming the tubules were pyknotic nuclei (Figure 1C). However, kidneys from BBR + DOX-treated rats showed much improvement in the glomeruli, glomerular membranes, and tubules to almost normal structure with very few degenerations. Most nuclei

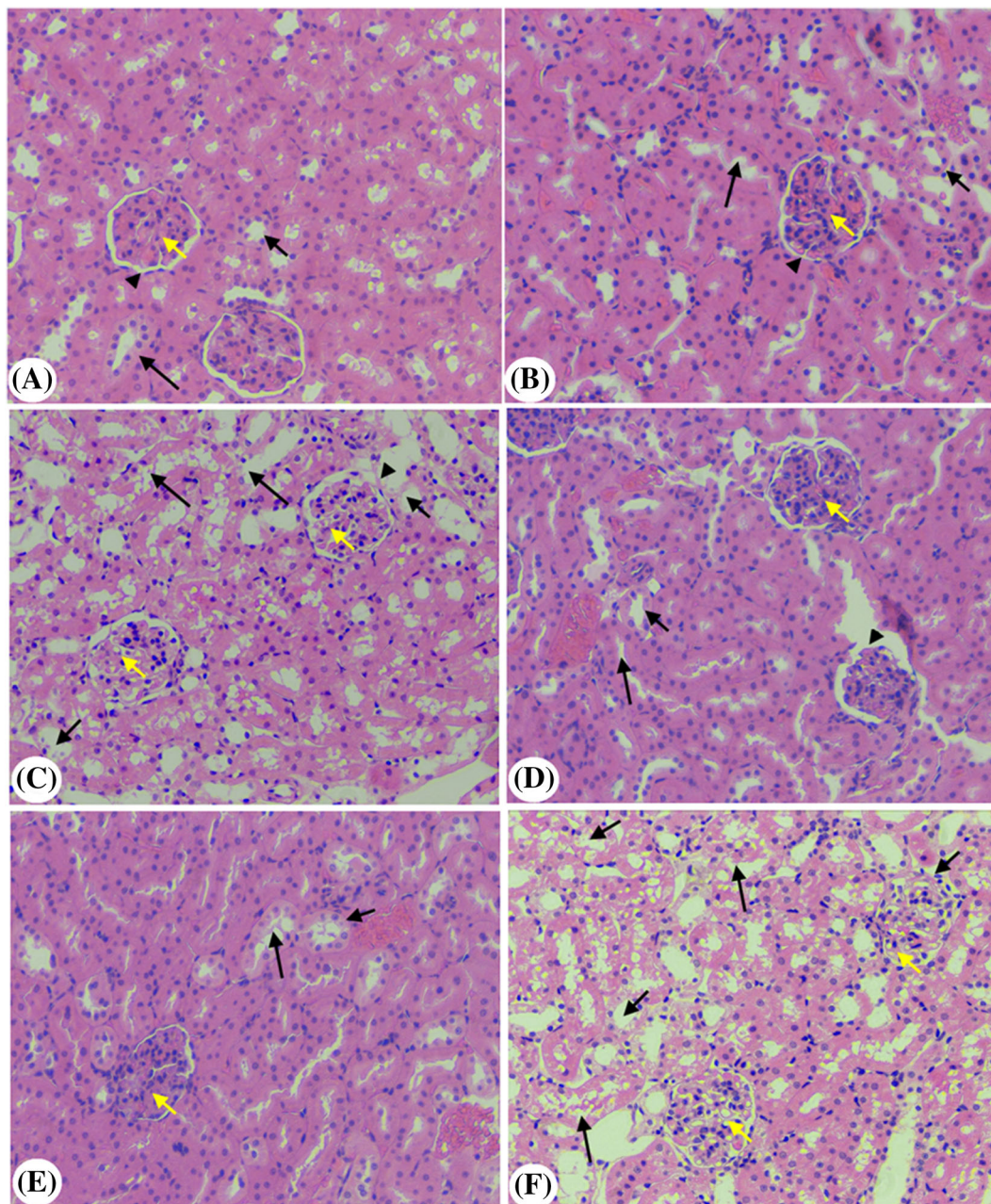
forming the tubules appeared (Figure 1D,E). Similar histological abnormalities in DOX-treated rats were also observed in BBR + DOX + brusatol-treated rats (Figure 1F).

### 3.3 | Changes in the expression and nuclear levels of Nrf2

Total renal cytoplasmic levels of Keap1 were not significantly altered between all groups included in this study (Figure 2A). mRNA, total cytoplasmic levels, and nuclear protein levels of Nrf2 were significantly decreased in the kidneys of DOX-treated animals (Figure 2B–D). On the other hand, mRNA, total cytoplasmic levels, and nuclear protein levels of Nrf2 were significantly increased in the kidneys of both the BBR and BBR + DOX-treated animals compared to the control DOX-treated rats, respectively (Figure 2B–D). mRNA, total cytoplasmic levels, and nuclear protein levels of Nrf2 have significantly decreased in the kidneys of BBR + DOX + brusatol-treated animals as compared to BBR + DOX-treated animals and were not significantly different from those measured in DOX-treated rats (Figure 2B–D).

### 3.4 | Changes in oxidative stress markers

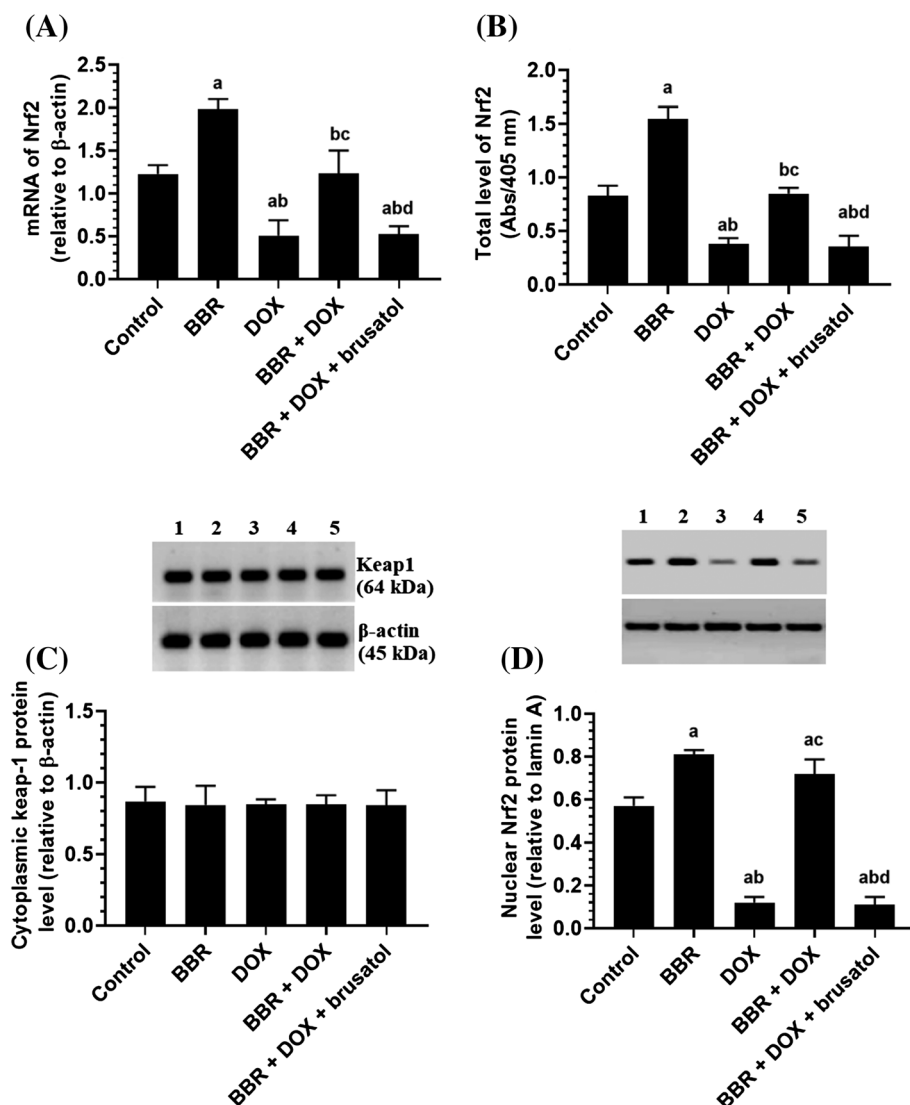
A significant increment in the levels of MDA and ROS with a concomitant reduction in the total levels of GSH



**FIGURE 1** Histological photomicrograph of the kidney sections of all groups of rats. (A) and (B) were taken from control and BBR-treated rats and showed normal histological features with intact glomerular (yellow arrow) and glomerular membrane (arrowhead), proximal tubules (PCT) (short arrow), and distal convoluted tubules (DCT) (long arrow). Most of the nuclei forming the tubules were normal. (C) was taken from DOX-treated rats and showed vacuolated glomeruli (yellow arrow), damaged glomerular membranes (arrowhead), PCT (short arrow), and DCT (long arrow). Note that most of the tubular nuclei were deeply stained and pyknotic (yellow arrow). (D) and (E) were taken from BBR + DOX-treated rats and showed the structure of the glomeruli with no vacuolization (yellow arrow) and glomerular membrane and with intact tubular structure in the majority of the PCT and DCT with normal nuclei (short and long arrows, respectively). However, some fields also showed some damage to glomerular membranes (arrowhead). (F) was taken from BBR + DOX + brusatol and showed similar abnormalities to those seen in group C.

and SOD was seen in the kidneys of DOX-treated rats as compared to control rats (Figure 3A–D). A significant increase in the total levels of GSH and SOD associated with a significant decrease in the levels of MDA and ROS were seen in the kidney samples obtained from

BBR and BBR + DOX-treated rats compared to control and DOX-treated groups (Figure 3A–D). Similar levels of these antioxidant markers were seen between the DOX and BBR + DOX + brusatol-treated rats (Figure 3A–D).



**FIGURE 2** The effect of all treatments on the expression and function of Keap-1/Nrf2 axis in the kidneys of DOX-treated rats. Data are given as mean  $\pm$  SD ( $n = 6$  animals/group); <sup>a</sup>vs. control rats; <sup>b</sup>vs. BBR-treated rats; <sup>c</sup>vs. DOX-treated rats; and <sup>d</sup>vs. BBR + DOX-treated rats

### 3.5 | Changes in inflammatory markers

Levels of all measured inflammatory markers were not significantly varied between the control and BBR-treated rats (Figure 4A–D). Kidneys of DOX and BBR + DOX + brusatol-treated rats showed a significant increase in the total levels of IL-6 and TNF- $\alpha$ . They had higher total nuclear protein levels and localization of NF- $\kappa$ B p65, which were significantly higher than their corresponding levels measured in the control or BBR + DOX-treated rats, respectively (Figure 4A–D). No significant differences in the levels of all these markers were depicted between DOX and BBR + DOX + brusatol-treated rats.

### 3.6 | Changes in apoptotic/antiapoptotic markers

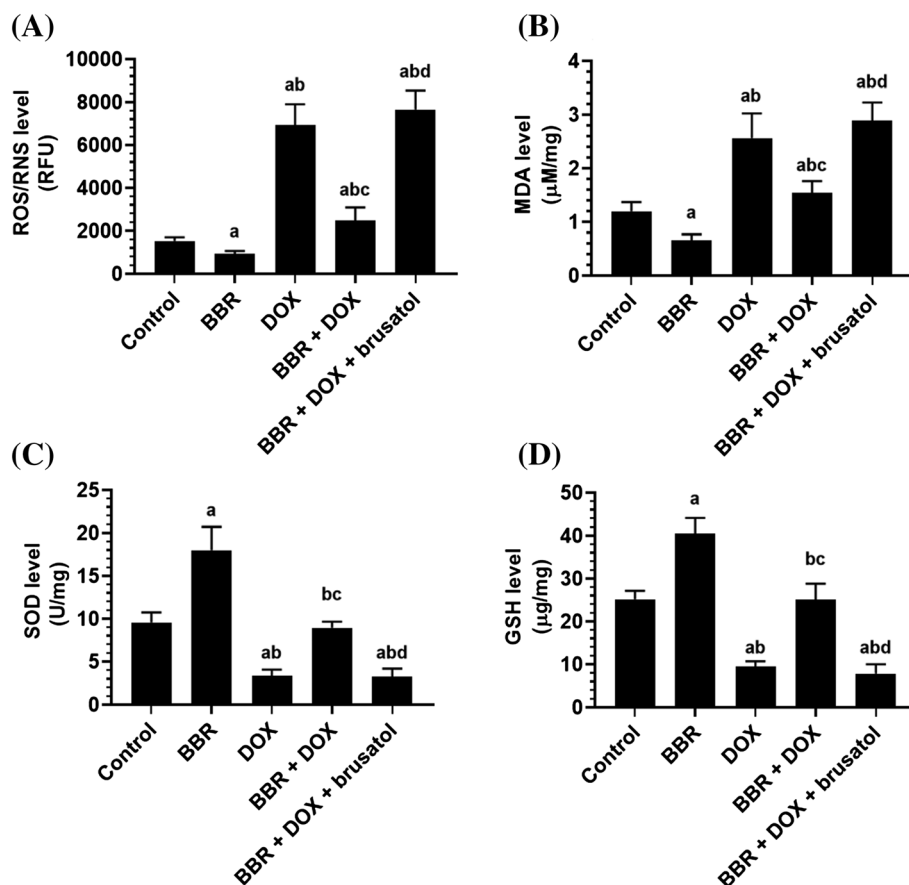
Only BCL2 protein levels were significantly increased in the kidneys of BBR-treated rats compared to the control

(Figure 5A). mRNA levels of Bax and caspase-3 and the ratio of Bax/Bcl2 were significantly increased in the kidneys of DOX-treated rats and were reversed with the administration of BBR (DOX + BBR) (Figure 5B–D). mRNA levels of Bcl2 were significantly lower, and mRNA levels of Bax and caspase-3, as well as Bax/Bcl2 ratio, were significantly higher in the kidneys of BBR + DOX + brusatol-treated rats as compared to BBR + DOX-treated rats (Figure 5B–D).

## 4 | DISCUSSION

The data obtained from this study support the limited existing evidence on the nephroprotective afforded by BBR. It also supports the studies of others who have demonstrated antioxidant neural and hepatic protective potentials of BBR in DOX rat models.<sup>44,45</sup> However, the data obtained from our current study also extended these data and showed that transactivation of Nrf2 through

**FIGURE 3** The effect of all treatments on selected markers oxidative stress/antioxidants in the kidneys of DOX-treated rats. Data are given as mean  $\pm$  SD ( $n = 8$  animals/group); <sup>a</sup>vs. control rats; <sup>b</sup>vs. BBR-treated rats; <sup>c</sup>vs. DOX-treated rats; and <sup>d</sup>vs. BBR + DOX-treated rats. ROS/RNS: reactive oxygen and nitrogen species, MDA: malondialdehyde, SOD: superoxide dismutase, and GSH: total glutathione



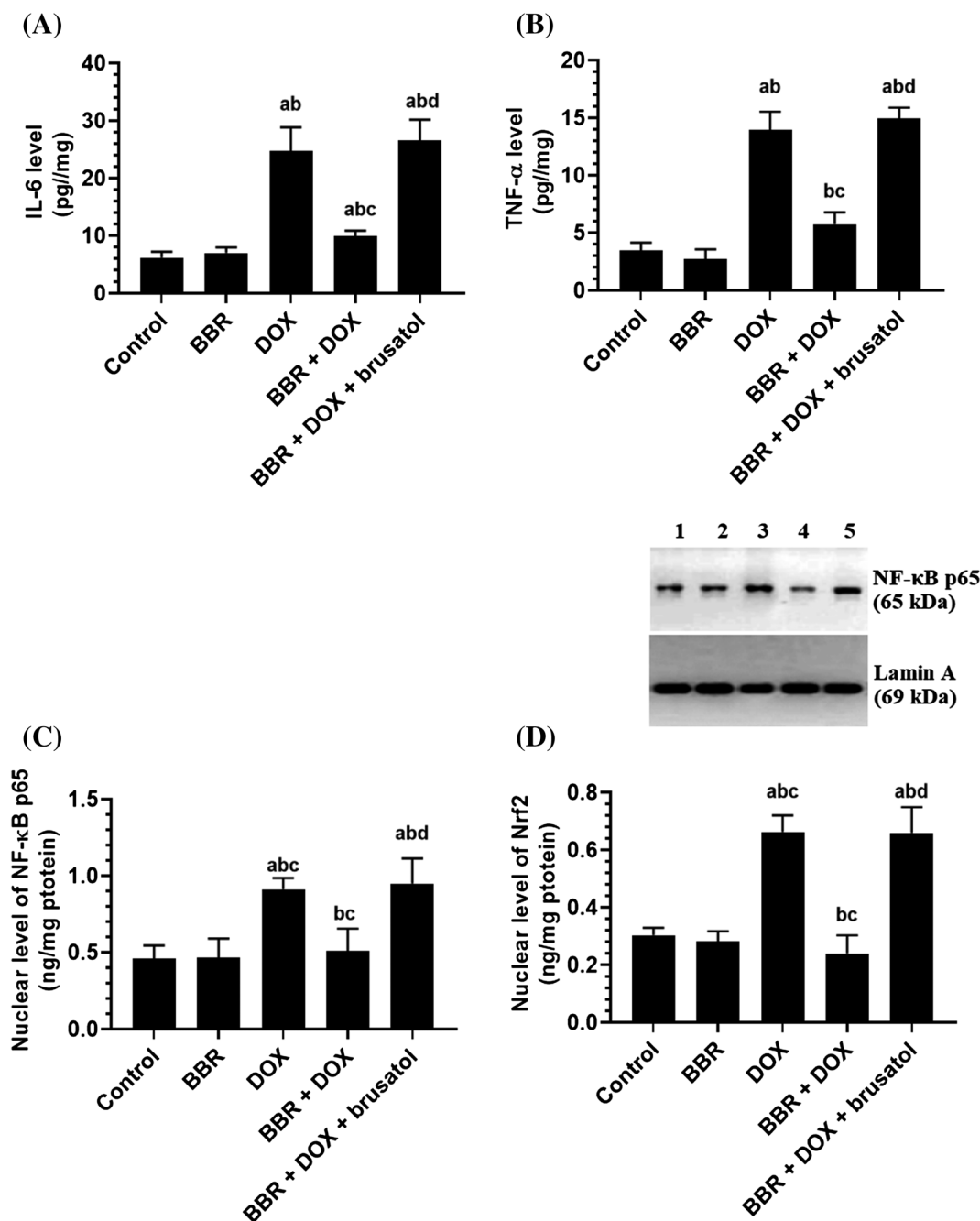
stimulating its transcription and inhibiting its cytoplasmic degradation through downregulating keap-1 expression is indispensable for this protection. Further, suppressing Nrf2 by brusatol abolished all the observed renoprotective effects afforded by BBR.

In this study, the single dose of DOX-treatment resulted in typical signs of nephrotoxicity, including renal tubular and glomerular damage, impaired Cr clearance (a marker of eGFR), and induced hypoalbuminemia, an increment in BUN, and micro/macroalbuminuria. In addition, DOX significantly impaired rat appetite, reduced dietary food intake, and reduced rat body weight. These results are in line with many previous investigations showing similar adverse effects on body weights and kidney function.<sup>10,44,50–52</sup> It could also be possible that the observed reduction in final body weight is explained by the cardiac, renal, and hepatic toxicities induced by DOX.<sup>7</sup>

The significant improvement in the glomerulus and tubular structures and the amelioration in the above-mentioned biochemical blood and urinary renal-related markers function after BBR treatment was the solid evidence presented in this study that confirmed the neuroprotection of BBR. This is not strange with this compound as daily administration of BBR attenuated not

only renal damage in DOX-treated animals but also in other renal conditions induced by other injurious stimuli, including ischaemia, ischaemia/reperfusion, hyperglycaemia, streptozotocin, hypertension, high glucose, and toxins (i.e., diclofenac), and was mediated mainly by suppressing oxidative damage, inhibiting inflammation, and increasing levels of the endogenous glutathione (GSH) and antioxidant enzymes.<sup>31,35,38–41,53</sup>

DOX-induced, high quantities of ROS are crucial for the induction of renal damage and proteinuria by promoting renal inflammation, fibrosis, and apoptosis.<sup>12,13,54</sup> In this context, DOX generates ROS within the cells by upregulating scavenging antioxidants, activating ROS-generating enzymes, promoting mitochondria damage, and forming Fe+2-DOX adducts.<sup>14–17</sup> In addition, DOX-induced ROS is the prominent trigger that induces lipid peroxidation and inflammation by stimulating the transcription of several inflammatory cytokines and activation of NLRP3 inflammasome and NF- $\kappa$ B.<sup>13</sup> Indeed, ROS is a potent stimulator of NF- $\kappa$ B and inflammatory cytokines, which, once activated, can trigger a vicious cycle of ROS generation.<sup>53,55</sup> Furthermore, ROS induced by DOX can activate cell apoptosis by upregulating p53 and Bax, as well as by suppressing Bcl2.<sup>56</sup> All these observations were also observed in the renal tissues of DOX rats in this



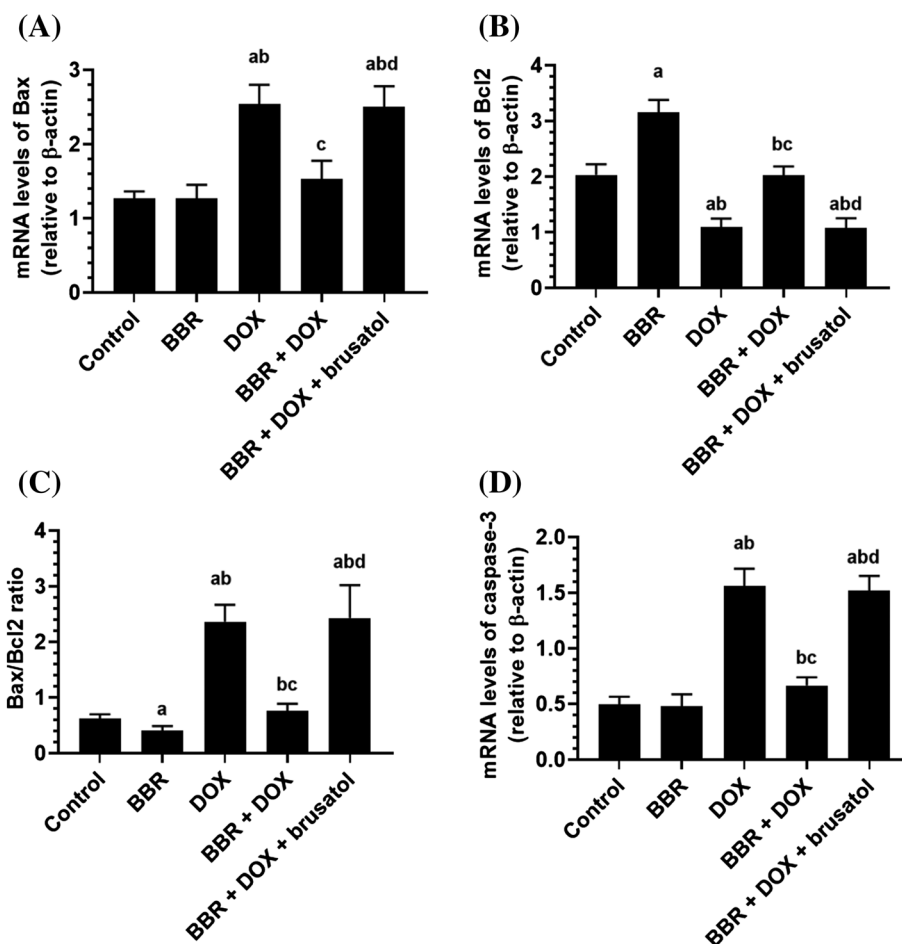
**FIGURE 4** The effect of all treatments on the levels of some inflammatory markers in the kidneys of DOX-treated rats. Data are given as mean  $\pm$  SD ( $n = 6$  animals/group); <sup>a</sup>vs. control rats; <sup>b</sup>vs. BBR-treated rats; <sup>c</sup>vs. DOX-treated rats; and <sup>d</sup>vs. BBR + DOX-treated rats. IL-6: interleukin-6, TNF: tumour necrosis factor

study, thus confirming that DOX renal toxicity is mediated by activating oxidative stress, inflammation, and intrinsic cells apoptosis.

On the other side, this study also confirmed that the renoprotection afforded by BBR is mediated, at least, by boosting antioxidants, inhibiting ROS, upregulating Bcl2, and suppressing all other damaging pathways (i.e., nuclear activation NF-κB, Bax, and caspase-3). However, since the administration of BBR in control

rats also significantly upregulated antioxidants and stimulated the transcription of Bcl2, without modulating nuclear levels of NF-κB neither the total content of TNF-α and IL-6 nor the mRNA expression of the apoptotic markers (i.e., Bax and Bcl2). Therefore, we concluded that the upregulation of endogenous antioxidants is a major common mechanism that underlies the anti-inflammatory and antiapoptotic nephroprotection of BBR.

**FIGURE 5** The effect of all treatments on the transcription of some selected markers of apoptosis. Data are given as mean  $\pm$  SD ( $n = 6$  animals/group); <sup>a</sup>vs. control rats; <sup>b</sup>vs. BBR-treated rats; <sup>c</sup>vs. DOX-treated rats; and <sup>d</sup>vs. BBR + DOX-treated rats



These data are in the same direction as those previously reported by Fouad et al.,<sup>44</sup> who have also shown that the nephroprotection afforded by BBR is mediated by stimulating levels and activities of numerous protective antioxidant enzymes (i.e., catalase and SOD) and the concomitant reduction in lipid peroxides. Also, other studies were conducted in different animal models to examine the nephroprotective effect of BBR and support that boosting antioxidants is the major mechanism of protection. In this view, BBR prevented DM-induced renal cell pyroptosis by decreasing the expression of NRLP3 inflammasome and caspase-3, as well as lowering levels of numerous inflammatory mediators in antioxidant-related mechanisms that involved stimulating MnSOD.<sup>41</sup> In the same line, BBR ameliorated gentamycin-induced nephrotoxicity and apoptosis in rats by increasing GSH and SOD levels, as well as stimulating Bcl2.<sup>57</sup> Also, BBR attenuated diclofenac-mediated renal damage in rats by stimulating SOD and glutathione reductase (GHSR).<sup>35</sup> It also prevented lead-induced kidney damage and apoptosis by boosting levels of GSH, GPx, SOD, and catalase.<sup>58</sup> Furthermore, the nephroprotective effect of BBR renal damage in high-salt-fructose-induced hypertensive rats was also associated with higher

content of these antioxidant markers.<sup>59</sup> However, searching the literature can also reveal tens of these studies, which can be listed here.

Nevertheless, recent evidence has revealed that DOX-associated renal damage is mediated and exaggerated by suppressing mRNA, protein, and nuclear translocation of Nrf2, which may explain the huge reduction in antioxidant levels.<sup>27–29</sup> Other studies have also shown that suppression of Nrf2 mediates DOX toxicity in other tissues, such as the liver and the heart.<sup>18,20,21,60,61</sup> This has been confirmed further in these studies, where we have reported similar results. On the other hand, the new finding concerning the BBR mechanism of protection against DOX-mediated renal injury in this study is that it is mediated by upregulation and activation of Nrf2. In this study, administration of BBR to control and DOX-treated rats significantly enhanced the mRNA and protein levels of Nrf2 with no effect on keap-1 expression, which typically stimulates Nrf2 cytoplasmic degradation. In addition, it increased the total cytoplasmic and nuclear protein levels of Nrf2 in the kidneys of the control and DOX-treated rats, which increased Nrf2 cytoplasmic levels could explain. This further illustrates why the kidneys of the control and DOX-treated rats had increased antioxidants

(i.e., SOD and GSH) and Bcl2 levels. Indeed, and in addition to its antioxidant potential, Nrf2 is a potent antiapoptotic factor that may inhibit apoptosis by upregulation of Bcl2. Also, we are demonstrating that upregulation and activation of Nrf2 is the major mechanism underlying BBR antioxidant, anti-inflammatory, and antiapoptotic effects. This has been confirmed by the observation that suppressing Nrf2 nuclear translocation by brusatol, prevented all renal protective effects seen after BBR administration.

Therefore, these data suggest that BBR not only modifies Nrf2 by downregulating keap1 but also by stimulating its transcription and translation. Although this is new to be shown in the kidneys of DOX-treated rats, the stimulation of Nrf2 by BBR was also demonstrated in vitro and in other tissues of different animal models. In support, BBR prevented high glucose-induced neural and renal toxicities in cultured cells by stimulating the Nrf2 and concomitant upregulation of HO-1, GSH, and SOD antioxidant components.<sup>42,62</sup> In the same line, Nrf2 prevented renal inflammation and pyroptosis in diabetic golden hamsters by upregulating and transactivating Nrf2.<sup>41</sup> BBR also attenuated hepatic damage in NAFLD and CCL4-treated rats by activating Nrf2/GSH/SOD axis.<sup>32,34</sup> However, despite these findings, our data can still determine the precise mechanism by which BBR induces its effect on Nrf2. Further studies may be required to target regulatory pathways such as SIRT1 and PI3K/Akt.

## 5 | CONCLUSIONS

The overall result of this experimental animal study suggests that BBR may be a novel target to alleviate DOX-induced renal damage by boosting the renal antioxidants and mainly through direct activation of Nrf2. However, this remains an observational study that requires further preclinical trials to confirm this. Such novelty may provide hope for cancer patients who suffer from such complications.

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## CONFLICTS OF INTEREST

The authors declare no potential conflict of interest with respect to the research, authorship, and/or publication of this article.

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